

Dust-Induced Light Loss (DILL): An IoT System for Monitoring PV Panel Dust

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Abstract—Dust accumulation on photovoltaic (PV) panels reduces sunlight absorption and significantly lowers energy output. Existing dust detection methods, such as PV output analysis, are often complex and data-intensive, and AI-based prediction, which is influenced by weather and climate conditions, necessitates location-specific datasets and fine-tuning to accommodate diverse environments. This situation creates a gap for simple, adaptable solutions suitable for small-scale applications, particularly in developing countries like Bangladesh. This study proposes a Dust-Induced Light Loss (DILL) detection system based on light sensors, which is simpler in design and operation and adaptable to various conditions. The system utilizes two Internet of Things (IoT)-enabled light sensors, one placed in an open area and the other under low-iron glass, to measure the intensity of visible sunlight (V_{SL}). By comparing the two readings, the system determines the V_{SL} loss through the glass caused by dust reflection, with a greater difference indicating more dust accumulation. Field testing shows that the system achieves 98.75% accuracy in dust detection and relies solely on sunlight, making it less susceptible to other environmental factors. It also supports remote monitoring via a web dashboard, enabling users to track dust accumulation in real-time. With accurate dust detection and remote access, this system enables users to determine a cleaning schedule, thereby solving barriers and facilitating the broader adoption of solar energy use.

Keywords—Dust-induced light loss (DILL), photovoltaic (PV) dust detection, IoT-enabled solar monitoring, sustainable monitoring, small-scale PV maintenance, climate-adaptive system.

I. INTRODUCTION

Recently, reliance on solar energy has grown as countries build PV power plants and expand use in rural areas to reduce future non-renewable resource shortages. In 2019, global solar PV installations totaled 111 GW, a significant increase from 29.5 GW in 2012. By the end of 2019, cumulative PV capacity had reached 623 GW [1]. The International Renewable Energy Agency (IRENA) set an ambitious goal with “The Remap Case 2050,” targeting an 86% energy use from renewable energy sources by 2050 [2]. As reliance on renewable sources worldwide increases, especially solar, overcoming the current obstacles is crucial for improving sustainability.

Dust buildup reduces PV panel efficiency, and the impact worsens if the panels remain uncleaned for a long time.

According to studies, dust deposition can decrease solar panel power output by 30% to 40% [3], resulting in a performance drop of up to 98% [4]. According to Al Siyabi et al. (2021), soiling losses of 7.5% and 12.5% reduced monthly generation by 5.6% and 10.8%, respectively, for a 2 MWp PV system in Oman [5]. Another study reported a 12.47% decline in output power and an 11.86% drop in the efficiency of PV systems in Al Karak, southern Jordan [6]. Adinoyi and Said (2013) reported a 50% reduction in the power output of PV panels in Saudi Arabia due to dust accumulation over six months [7]. Therefore, dust buildup on PV panels is a significant factor in reducing performance, requiring regular cleaning to keep them working efficiently.

Although current dust detection approaches have made progress, they still face challenges in adaptability across varying locations and weather conditions. This highlights the need for developing methods that are less affected by environmental factors, simpler in design, and more cost-effective. The primary objectives of this study are to (i) design a robust and straightforward dust detection system that can withstand environmental variations, (ii) assess its performance in real field conditions and demonstrate its adaptability to different environments, and (iii) implement remote real-time monitoring of dust accumulation via an IoT-enabled web dashboard. The proposed DILL detection system utilizes an IoT-enabled light sensor positioned beneath a glass surface to measure V_{SL} intensity after passing through dust-accumulated glass. This setup mirrors the configuration of PV panels, where solar cells are located beneath a 2 mm thick glass layer. A second light sensor, placed in an unobstructed area, records the direct V_{SL} intensity. After measurement, both sensor readings are compared, and a lower V_{SL} intensity under the glass compared to the open space indicates V_{SL} reflected from the glass due to dust. The larger the difference, the greater the dust accumulation on the surface, triggering the initiation of the cleaning process.

This study explores the development of an IoT-based dust condition monitoring system for PV panels, designed to enable real-time remote monitoring. It investigates the reduction in V_{SL} intensity caused by naturally deposited dust on glass surfaces, analyzing V_{SL} blockage over time. It also evaluates weekly and monthly trends in dust accumulation and their impact on V_{SL} intensity reduction. Beyond just detecting dust, this study defines different dust levels based on V_{SL}

blockage, providing users with a clear indication of the panels' condition. The study also assesses the proposed system's accuracy, compares it with recent approaches, demonstrates its potential adaptability to diverse environments, and suggests optimal operational conditions.

II. LITERATURE REVIEW

Olorunfemi et al. (2022) developed a method for detecting dirt from solar panels using Arduino and a TCS3200 color sensor. The system identified dirt on the solar panel's surface with RGB color values and started the cleaning mechanism to remove it [8]. The color sensor has an effective detection range of only 3cm. As the distance between the sensor and the solar panel increases, the accuracy of dust detection declines.

Shao et al. (2024) introduced an improved Adam optimization algorithm designed for detecting dust. The traditional Adam algorithm was upgraded to improve detection accuracy, stabilize the learning rate during training, and avoid problems such as local optima and overfitting. The model enabled faster identification and accurate dust detection on the surface of PV panels, achieving up to 99% accuracy [9]. However, deep learning models require vast amounts of historical data for practical training, which increases the overall cost and complexity.

Bashir et al. (2025) developed a combined convolutional neural network (CNN) and random forest (RF) method to detect dust buildup on PV panels in dry regions. The method transforms electrical I-V curve data, covering voltage, current, and power, into vibrant RGB mosaic images, which helps make the identification of dust buildup much more accurate. The hybrid CNN-RF model achieved outstanding results, reaching 98% accuracy based on the image data for classifying dust severity [10]. One limitation of the method is that it has only been tested on small datasets and in limited weather conditions, which makes it harder to know how well it would work with larger data or in real-time use for maintenance.

Bassil et al. (2025) used fine-tuned deep learning models to detect dust on solar panels and passed the extracted features to simple tree-based classifiers. The approach involved extracting visual features from solar panel images using fine-tuned EfficientNetB7 and vision transformer models, then feeding these features into tree-based classifiers to achieve accurate dust detection under diverse environmental conditions, reaching 97% accuracy [11]. However, the proposed method has limitations due to the small dataset, which may not accurately reflect real-world conditions, such as different dust types, lighting, or panel angles.

The literature review summarizes that numerous studies have been conducted to detect dust on PV panels, with some achieving higher accuracy. However, some issues in the existing system make it complex, challenging to maintain. Machine learning (ML) is a widely used approach that has shown high accuracy and strong performance in outdoor PV panel monitoring, analyzing captured images of panels to detect dust accumulation. A limitation arises when the ML model requires historical photos or data from the specific site for practical training, as such data are not always available for every location. Additionally, extreme or unusual weather events, such as sudden heatwaves, unseasonal rainfall, fog, or storms, can affect image quality and detection accuracy. To maintain performance under these conditions, the model often requires fine-tuning and, in some cases, full retraining when

deployed in new locations or under different weather patterns. This method often relies on a high-resolution camera and, in many cases, an Unmanned Aerial Vehicle (UAV) to capture the entire PV area, which increases both costs and maintenance requirements. Moreover, dust detection using PV system data involves analyzing multiple parameters, such as solar irradiance, voltage, current, and temperature, to identify performance drops and predict dust-related losses. In previous studies, a light sensor has not been utilized to define the presence of dust on PV. By considering these issues, an adaptable dust detection system has been developed to operate in various environmental conditions and a changing climate, enabling easy real-time monitoring. In addition, an IoT-enabled light sensor costs around \$6, which also makes the proposed system cost-effective.

III. POTENTIAL OF IoT AND SENSOR-BASED SOLUTIONS IN SOLAR ENERGY APPLICATIONS

Ramadhan et al. (2021) implemented and tested an IoT-based monitoring system for an off-grid solar power plant by utilizing ESP32 as a microcontroller, INA219 and PZEM004T sensors for data collection, the MQTT communication protocol to transmit data to a Google cloud platform, and store it in a MySQL database [12]. Putra et al. (2021) developed a portable solar power plant integrated with an IoT-based energy management system. Voltage, current, power usage, and battery capacity were continuously measured via sensors, processed by an Arduino ATmega328, and transmitted through an ESP8266 Wi-Fi module to the cloud, enabling access from smartphones, tablets, or computers remotely [13]. Spanias (2017) introduced IoT-enabled Smart Monitoring Devices (SMDs) for PV panels to transform them into programmable IoT nodes. Each SMD integrates voltage, current, temperature, irradiance sensors, radios, and relays to support real-time panel monitoring, fault detection, dynamic topology reconfiguration, machine learning, and computer vision-based shading prediction, as well as remote control [14]. Gochhait et al. (2022) demonstrated how IoT can enhance and simplify the management of solar energy systems. They utilized an Arduino-based setup and sensors to monitor solar panels in real time, checking their performance, conditions, and power output. The information is sent to the cloud so it can be viewed from anywhere. With IoT, the system can move panels to follow the sun, detect problems early, and maintain high performance without requiring on-site personnel [15]. Advanced sensor technologies enable precise, real-time data collection, thereby enhancing accuracy and reducing the need for human intervention. Similarly, IoT solutions facilitate real-time monitoring and remote management, significantly improving operational efficiency. This study combined these technologies to build a dust detection system that is simple and easily accessible to everyone.

In this framework, IoT connects the physical sensors to the cloud, allowing users to track changes in dust levels remotely, without requiring a physical presence near the panel. The IoT component is implemented through the integration of an ESP-32 microcontroller, which reads sensor data, processes it, and transmits the results to a cloud-based platform (Blynk) for real-time access.

IV. BASIC IDEA OF THE PROJECT

Fig. 1 illustrates the innovative design of the dust accumulation measurement, which enables the definition of

dust levels and informs the user about the condition. This method involves placing a clear, low-iron glass panel next to the PV panels. When dust settles on the transparent glass, it blocks some V_{SL} . An IoT-enabled light sensor is mounted beneath the glass, with another sensor outdoors, controlled by a servo motor to move up and down. The DSServo DS3218 motor provides precise control over the sensor's angle movement, ensuring accurate measurements of sunlight. Besides, the motor ensures that no dust accumulates on the open-space glass by keeping it facing downward at all times. The light sensor in the open area measures direct V_{SL} from the sun, while the second sensor, located under glass, measures the transmitted V_{SL} . These readings are then compared to find the difference. When there is heavy dust on the glass, the open sensor shows a much higher lux value than the sensor under the glass because the dust reflects more V_{SL} . For a clean glass surface, both sensors show almost the same lux value, meaning very little V_{SL} is lost. Measuring V_{SL} intensity under the glass also defines glass transparency, and lower glass transparency defines a high level of dust above it. Sensor data and dust levels are transmitted to the cloud via WebSockets for storage, and are visualized in real-time on the web dashboard and mobile app for remote monitoring.



Fig. 1. Proposed IoT-enabled dust detection system comparing sunlight intensity through glass and open air.

V. HARDWARE IMPLEMENTATION OF DILL DETECTION SYSTEM

The proposed dust detection system features a straightforward hardware design comprising sensors, a servo motor, a sensor node, and low-iron transparent glass. The sensor node primarily includes a microcontroller (ESP-32), a buck converter, a printed circuit board (PCB) to house all electronic components, and an enclosure to protect the circuitry. Fig. 2 illustrates the 3D design of the proposed system as implemented under field conditions. The 3D design shows transparent glass is cut to the same length as the PV panels and positioned directly beside them, allowing both to be cleaned simultaneously during washing. Fig. 3 presents the small-scale prototype setup of the system for data collection, replicating the 3D design.

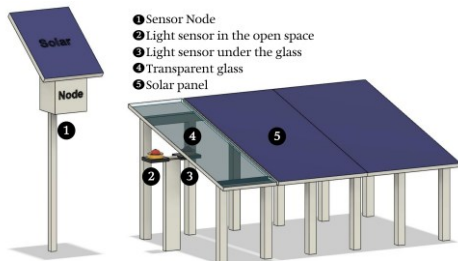


Fig. 2. 3D hardware design of the dust detection system with sensors and glass placement.

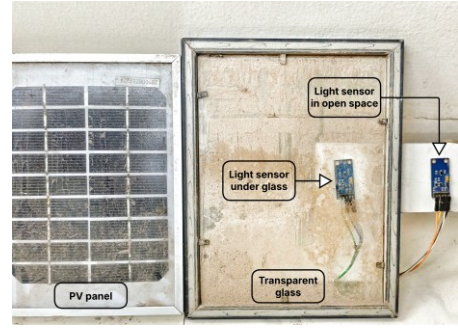


Fig. 3. Prototype setup replicating the 3D design for field data collection.

The ESP-32 manages the operations, including data acquisition, processing, sending data to the server, and receiving commands from it. The firmware running on the ESP-32 acts as the central processing unit for the IoT-based dust detection system. Ambient light sensors communicate with the ESP-32 microcontroller via the Inter-Integrated Circuit (I²C) protocol to measure light intensity both through dusty glass surfaces and in open space. The system synchronizes time using an NTP server and performs measurements daily at noon. It calculates the percentage reduction in light intensity using a relative change formula. Based on the calculated dust levels, the system triggers predefined events and sends the results to the Blynk IoT platform, where they are displayed on the web dashboard or mobile app. The entire operation of the dust detection system is outlined in the pseudo code in Table I.

TABLE I. ALGORITHM OF THE DILL-BASED DETECTION SYSTEM

Pseudo-code
<pre> 01 // Pseudo code for IoT-based monitoring system 02 DEFINE libraries: WiFi, Blynk, I2C, Time, Servo 03 DEFINE WiFi credentials, Blynk token, and light sensor address 04 INITIALIZE variables: Clean_Glass_Light, Dirty_Glass_Light 05 START Serial communication (115200 baud) 06 INITIALIZE I2C for light sensor 07 CONNECT to WiFi (restart ESP32 if fails) 08 CONNECT to Blynk 09 SYNC time with NTP server 10 // Function to Get Current Time 11 FETCH local time from system clock 12 RETURN current hour (or -1 if failed) 13 // Sensor Reading Function 14 ROTATE servo 180° forward (to position sensor face upper side) 15 REQUEST light intensity measurement from light 16 READ sensor data (Clean & Dirty Glass Light) 17 CONVERT raw data to lux (using sensor formula) 18 ROTATE servo 180° reverse (back to original position) 19 PRINT both light intensity values 20 // Data Processing & Transmission 21 COMPUTE Relative Change: 22 ABS(Dirty_Glass_Light - Clean_Glass_Light) / Clean_Glass_Light * 100 23 TRIGGER Blynk events based on dust levels: 24 IF RelativeChange < 10 → "low_dust" 25 ELSE IF 10 ≤ RelativeChange ≤ 20 → "moderate_dust" 26 ELSE IF 20 < RelativeChange ≤ 30 → "high_dust" 27 ELSE → "extreme_dust" 28 SEND data to Blynk and PRINT "Data Sent Successfully" 29 // Main Loop Execution 30 RUN Blynk process 31 GET current hour 32 IF current hour == 12: 33 CALL Sensor Reading Function 34 CALL Data Processing & Transmission Function 35 // Helper Function to Write Data to Sensor 36 SEND configuration byte to sensor via I2C </pre>

A. Working Principles of the DILL-based Detection System

The system's working principles are outlined in Fig. 4. The system performs measurements daily at noon, as sunlight intensity is generally stable during this period, reducing variability in data collection. The microcontroller reads the light sensor values for the open space and the area under the glass. Then, both values are compared to determine the reflected and absorbed V_{SL} , which in turn defines dust levels. The measured data is transmitted to the Blynk cloud and mobile application via IoT, where it is displayed in real-time. A higher level of reflected V_{SL} indicates greater dust accumulation on the PV panel, triggering timely alerts for cleaning. When dust levels remain low, no notifications are issued, avoiding unnecessary maintenance actions.

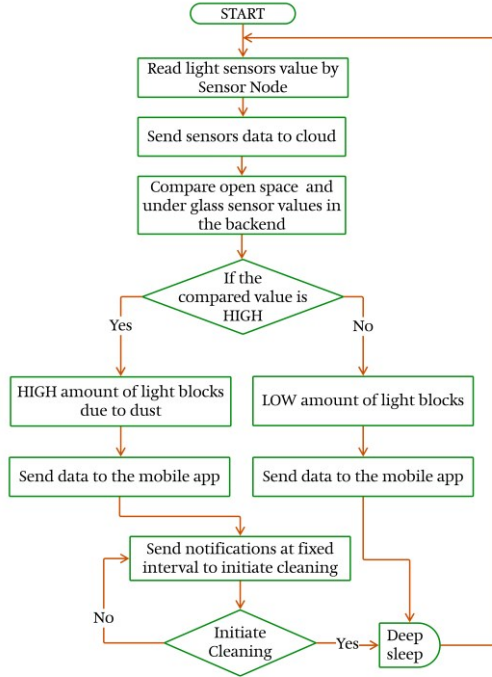


Fig. 4. Flow chart of the DILL system showing data collection, processing, and transmission.

The relative change formula, Equation 1, quantifies dust accumulation by comparing the light intensity measured under the glass (with dust) to the light intensity measured in open space (without dust). This formula calculates the percentage reduction in sunlight intensity to determine the relative difference between the two measurements. Since both measurements are taken simultaneously, any fluctuations in sunlight intensity affect both sensors equally, thereby normalizing the data and making dust detection independent of changing sunlight conditions. Here, $L_{under_clean} = V_{SL}$ in open space, $L_{under_dust} = V_{SL}$ under glass.

$$Relative\ Change = \frac{L_{under_dust} - L_{under_clean}}{L_{under_clean}} \times 100 \quad (1)$$

B. Uncertainty and Error Analysis

Standard uncertainty estimates how much a measurement might vary due to random error, covering a 68% confidence interval. As an instrument's accuracy range is known, assuming the errors are spread evenly across that range, the accuracy is divided by $\sqrt{3}$ to find the standard uncertainty [16]. Expanded uncertainty is achieved by multiplying the

standard uncertainty by a factor of 2 to cover a 95% confidence interval. Equation (2) indicates the uncertainties and errors caused by using instruments over the experimental procedures, where u is the standard uncertainty and a is the instrument's accuracy.

$$u = \frac{a}{\sqrt{3}} \quad (2)$$

Uncertainty and error estimates were checked in the field by comparing the system's readings with those from a handheld lux meter. As the lux meter also carries a margin of error, this uncertainty was taken into account in the analysis to make sure the validation remained unbiased. During data collection, an error of around 5% was consistently observed throughout the entire experimental process, as shown in Table II.

TABLE II. ACCURACY, STANDARD UNCERTAINTY, AND ERROR OF THE MEASURING DEVICE

Instruments	Accuracy	Range	High. measuring point	Standard uncertainty	Expanded uncertainty	Error (%)
*Light Sensor	$\pm 4\%$	1- 88,000 lux	80,000 lux	1848 lux	3696 lux	4.62%
Lux Meter	$\pm 5\%$ ± 10	1- 200,000 lux	80,000 lux	2317 lux	4633 lux	5.79%

*The TSL2591 is a high-sensitivity digital light sensor with a wide dynamic range, capable of accurately measuring ambient light.

VI. RESULTS AND DISCUSSIONS

A. Remote monitoring through the web dashboard

Fig. 5 shows the Blynk web dashboard's data visualization system, which enables remote monitoring of light sensor data. Both light sensors' data and computed V_{SL} are visualized using three gauges for better understanding. Blynk Web Dashboard is an easily customizable platform that enables users to create data streams, events, and notifications, and seamlessly connect the Blynk cloud with IoT devices. While the premium subscription offers additional features, the free plan is also functional, providing up to 30,000 data points per month.

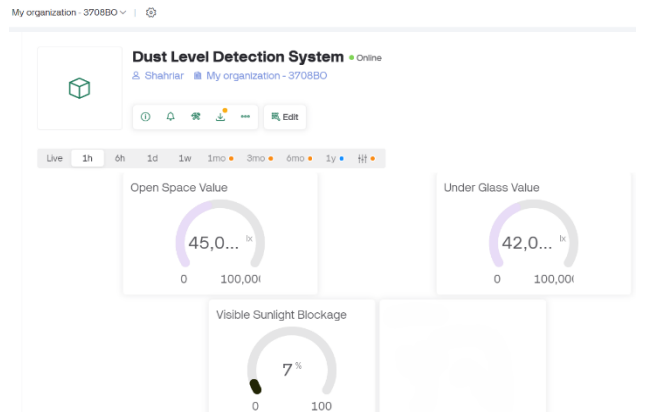


Fig. 5. Blynk dashboard visualizing real-time sensor readings and dust levels.

B. Variations in V_{SL} Intensity for Naturally Deposited Dust on Glass

Fig. 6a compares V_{SL} intensity measurements under clean glass and glass with naturally accumulated dust, shown as two separate bars. The blue bar represents the lux value of

V_{SL} passing through clean glass, and the red bar represents the amount of V_{SL} passing through dusty glass. The relative change formula is used to differentiate the V_{SL} transfer amount for both glasses. The graph in Fig. 6b illustrates the percentage decrease in V_{SL} intensity over time under the dusty glass. Within one month, the dust on the glass reduced the sunlight passing through from 2% at the start to approximately 34% at the end.

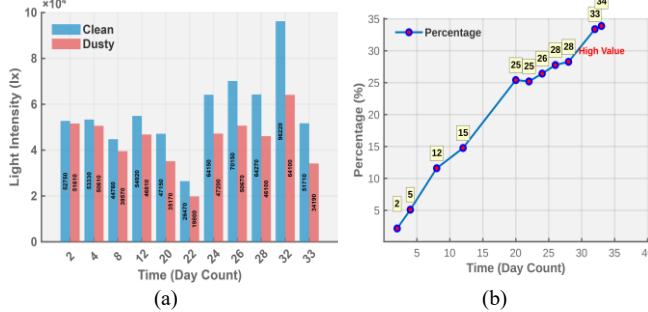


Fig. 6. (a) Visible sunlight intensity under clean and dusty glass over time, and (b) percentage reduction trend within one month.

C. Weekly and Monthly Scenarios of Naturally Deposited Dust on Glass

To quantify the decrease in V_{SL} intensity beneath the glass over time, data were recorded between 9 AM and 5 PM at three intervals of natural dust buildup: 1 week, 15 days, and 1 month. Fig. 7 shows the effect after one week, with the equation $y = 0.9112x$, meaning the intensity beneath the dusty glass is approximately 91.12% of the direct light intensity. Fig. 8, which represents 15 days of dust buildup, gives $y = 0.8017x$, indicating that transmission drops to about 80.17% as the dust increases. Fig. 9, showing one month of dust accumulation, gives $y = 0.6804x$, meaning only 68.04% of the light passes through, reflecting a significant reduction in transmission. Overall, these results highlight a gradual increase in dust accumulation, leading to a corresponding decrease in V_{SL} transmission.

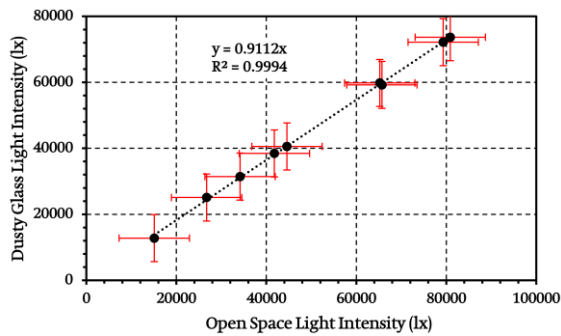


Fig. 7. Linear correlation between open space and dusty glass light intensity after one week of accumulation ($y = 0.9112x$, $R^2 = 0.9994$).

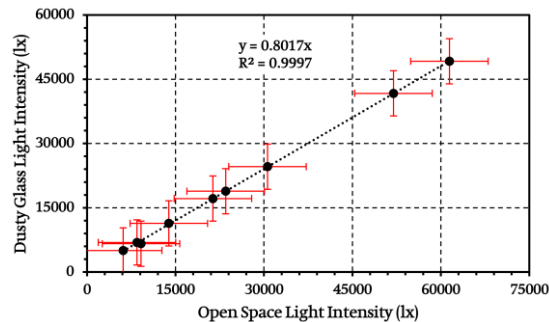


Fig. 8. Linear correlation between open space and dusty glass light intensity after fifteen days of accumulation ($y = 0.8017x$, $R^2 = 0.9997$).

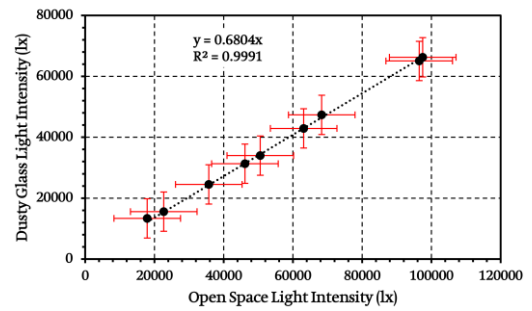


Fig. 9. Linear correlation between open space and dusty glass light intensity after one month of accumulation ($y = 0.6804x$, $R^2 = 0.9991$).

Table III defines the various dust levels based on field data. According to the experimental results, a 10–20% decrease in V_{SL} intensity corresponds to a moderate dust level, a 20–30% decrease indicates a high dust level, and a change greater than 30% signifies an extreme dust level on the PV.

TABLE III. DEFINING DUST LEVELS BASED ON EXPERIMENTAL DATA

* V_{SL} intensity change (%)	Dust level
0-10	Low
10-20	Moderate
20-30	High
> 30	Extreme

* V_{SL} intensity change calculated using Equation (1).

D. Detection accuracy of the proposed system

Data on V_{SL} intensity under dusty glass were also taken for four levels of artificially deposited dust by the air blow method. 0.10 g, 0.25 g, 0.38 g, and 0.50 g of dust were uniformly distributed on a 0.031 m² glass area, which refers to dust densities of 3.3 g/m², 8.06 g/m², 12.25 g/m², and 16.13 g/m², respectively. Each dust density corresponds to the following dust levels: Low (3.3 g/m²), Moderate (8 g/m²), High (12.25 g/m²), and Extreme (16.13 g/m²).

For each dust level, 20 measurements were recorded using both the handheld digital lux meter and the light sensor. The lux meter readings were considered as the actual dust levels, which were then compared with the levels detected by the proposed system. The results are summarized in Table IV, which presents the confusion matrix of detected versus actual dust levels for artificially deposited dust.

TABLE IV. CONFUSION MATRIX OF DETECTED VS. ACTUAL DUST LEVELS FOR ARTIFICIALLY DEPOSITED DUST

Actual \ predicted	Low	Mode rate	High	Extre me	Row total
Low	20	0	0	0	20
Moderate	0	20	0	0	20
High	0	0	19	1	20
Extreme	0	0	0	20	20
Column Total	20	20	19	21	80

From the 80 total observations, 79 were correctly classified, resulting in a detection accuracy of 98.75%. This result demonstrates that the proposed system can accurately detect both the presence and severity of dust without relying on PV performance data or reference power output values.

E. Comparative analysis with existing work

Table V compares selected existing sensor-based dust detection methods for PV panels in terms of detection accuracy. While sensor-based approaches have demonstrated

good capability in detecting dust, quantitative validation is often missing. This study contributes a quantitative, accuracy-validated sensor-based PV dust detection system using a dual-sensor DILL approach. Field testing achieved 98.75% detection accuracy for defined dust levels without dependence on location-specific historical datasets. For this comparison, ML-based systems are excluded because their datasets, detection approaches, and evaluation protocols differ substantially from those of sensor-based methods, which can make direct comparison unreliable.

TABLE V. COMPARISON BETWEEN RECENT WORKS

Reference	Detection method	Accuracy
[8]	Color sensor-based dirt detection	Not reported
[17]	Sensor-based data acquisition combined with heuristic and statistical	94%
[18]	Dust sensor-based dust detection.	95%
This study	DILL-based dust detection	98.75%

F. Optimal operational condition of the proposed system

The intensity of sunlight varies throughout the day, which can affect the reliability of the system's measurements. Sunlight naturally fluctuates depending on meteorological conditions and the position of the sun. The time of day, whether morning, noon, or evening, has a significant influence on this phenomenon. For example, sunlight is generally most intense around noon, whereas its intensity diminishes in the early morning and late afternoon. To ensure precision, the system should collect data at a fixed time to minimize the effects of uniform light and solar angle variations. Additionally, regular sensor calibration ensures accuracy and use of advanced data processing, including noise reduction and smoothing, which filters out irregularities and enhances the reliability of dust detection.

G. Significance of the proposed method

The proposed dust detection system can be used on a small scale, as well as in large-scale implementations. Solar energy sources are virtually unlimited and easily accessible. Electricity generation from solar energy is straightforward, as seen with PV panels placed in open spaces to collect energy and connect to a battery for storage. People are increasingly installing this system in their households and on rooftops to ensure power backup during load shedding. The number of PV panel installations for personal home use is rapidly increasing. People use solar energy for their own needs and contribute to the local electricity grid through net-metered rooftop solar systems. According to the National Database of Renewable Energy in Bangladesh, approximately 117 MW of net-metered rooftop solar capacity has been installed [19]. Despite this large capacity, no effective dust detection system has been widely implemented due to the complexity of existing design, installation, and maintenance solutions. For small-scale solar systems, the complexity and high cost of dust detection systems discourage people from using them. As a result, such systems are still not commonly used in developing countries like Bangladesh. The proposed system is simple in design, requires less complex maintenance, offers high detection accuracy, facilitates easy remote monitoring, and is cost-effective, which can lead to widespread adoption.

The combined features of the proposed system enhance monitoring accuracy and potential adaptability under varying environmental conditions. This detection system, when integrated with a cleaning mechanism, can automate the entire process. This approach is particularly valuable for arid regions or large solar farms where frequent on-site monitoring and cleaning are challenging. The monitoring system is also highly effective for rooftop PV systems in urban areas due to its simple construction and ease of maintenance. It enables local users to easily and remotely identify the reasons behind declining PV efficiency and restore performance. Together, these features enhance the operational efficiency of PV monitoring and help promote the broader adoption of solar energy.

VII. ENVIRONMENTAL ADAPTABILITY POTENTIAL

Environmental factors, such as rain, storms, fog, and variations in sunlight, impact most computer vision-based dust detection techniques, often misleading models and requiring fine-tuning. However, the proposed system relies solely on sunlight, as it utilizes the dust-induced light loss method. Sunlight also varies by location and frequently changes throughout the day. This was one of the primary challenges during development, which was later solved by incorporating an extra light sensor placed openly, operating simultaneously with the other sensor. The open-space light sensor tracks variations in sunlight, helping to compare and determine the actual V_{SL} intensity reduction under the glass due to dust. By resolving this, the system can be implemented anywhere, as sunlight is globally accessible. The system has also been tested in two different locations, at various times of the day, under different sunlight conditions, and has demonstrated consistent performance, indicating its potential adaptability. The simple construction also minimizes the risk of failure in dusty, wet, or high-temperature environments.

VIII. LIMITATIONS

In this study, data were collected from Dhaka (23.88, 90.39) and Gazipur (23.98, 90.41), which are very close to each other; therefore, the results may not accurately reflect performance in widely different locations. Testing was also conducted at the end of summer and the start of the rainy season, which means other seasonal conditions were not covered. Future testing in different regions and during various seasons would help confirm the system's long-term reliability.

In real use, other things besides dust can also block sunlight from passing through. For example, bird droppings or shade from nearby objects can cause the readings to drop in a similar way. Over time, the glass surface may become scratched or worn, which can alter how it transmits light and make the measurements less accurate. These issues can sometimes cause the system to detect dust when none is present. So, periodic maintenance may be needed to distinguish dust from other light-blocking factors.

The proposed system requires some maintenance, including initial sensor calibration, occasional hardware replacement, and fixing software bugs or potential failures. While the initial calibration is essential for ensuring accurate readings, it is a one-time process.

IX. CONCLUSIONS AND FUTURE WORK

In this work, a novel DILL approach was developed for the automatic detection of dust in PV systems and tested for its performance under real-world conditions. The IoT-driven setup relies on an ambient light sensor positioned beneath transparent low-iron glass, where the reduction in V_{SL} intensity serves as an indicator of dust accumulation. The main outcomes of the study are outlined below: Dust accumulation on the PV panel was detected by measuring V_{SL} intensity under the glass with 98.75% accuracy. The data was sent to the Blynk cloud and displayed in the dashboard, enabling remote monitoring of dust accumulation on PV panels. The findings determined that dust levels range from low to extreme based on V_{SL} intensity reduction, with a drop of more than 30% indicating extreme dust accumulation. Moderate dust levels are observed when the V_{SL} intensity decreases by 10–20%. The proposed system relies primarily on sunlight and uses two sensors to monitor intensity changes, which makes the system adaptable to diverse environmental conditions.

Although the system already achieves 98.75% dust detection accuracy, further improvements could target scalability for larger PV setups. Enhancing efficiency, such as reducing power consumption and response times for remote monitoring, can optimize long-term operations, particularly in resource-constrained areas. In the future, testing across various regions and seasons is necessary to verify the system's long-term reliability. Future research will also evaluate the system's performance at night using a DC bulb as a controlled light source. The use of artificial light sources ensures stable illumination for measuring reflected light, which is essential for detecting dust.

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